



Sydney Girls High School

2006
TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics

Extension 1

This is a trial paper ONLY.
It does not necessarily
reflect the format or the
contents of the 2006 HSC
Examination Paper in this
subject.

General Instructions

- ♦ Reading Time – 5 mins
- ♦ Working Time – 2 hours
- ♦ Attempt ALL questions
- ♦ ALL questions are of equal value
- ♦ All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- ♦ Standard integrals are supplied
- ♦ Board-approved calculators may be used.
- ♦ Diagrams are not to scale
- ♦ Each question attempted should be started on a new sheet. Write on one side of the paper only.

Marks

Question 1 (12 marks)

- a) Differentiate $y = 4 \cos^{-1} 3x$ (2)
- b) Find the coordinates of the point that divides the interval between (3, 4) and (-5, 1) externally in the ratio 2:3 (2)
- c) Find $\int \frac{dx}{4+3x^2}$ (2)
- d) Sketch the graph of $y = \sin(\sin^{-1} x)$ showing clearly the domain and range (3)
- e) Find $\int 2xe^{x^2-3} dx$ using the substitution $u = x^2 - 3$ (3)

Marks

Question 2 (12 marks)

- a) Find the obtuse angle between the lines $y = 2x$ and $5x + 2y - 3 = 0$ to the nearest minute. (2)
- b) Sketch $y = \frac{|x|}{x^2}$ (2)
- c) Find the exact value of $\int_0^{\frac{\pi}{6}} \cot x dx$ (3)
- d) Find the general solution of $\tan 2\theta = \sqrt{3}$ (2)
- e) Find the equation of the normal to the curve $y = \sin^{-1} 3x$ at the point where $x = 0$ (3)

Marks

Question 3 (12 marks)

- a) The rate of change in velocity (v) of a falling object is given by $\frac{dv}{dt} = -k(v - C)$ where k and C are constants.
- (i) Show that $v = C + Ae^{-kt}$ is a solution of this equation (1)
- (ii) When $C = 500$, the initial velocity is 0 and the velocity (v) after 5 seconds is 21 ms^{-1} . Find A and k . (2)
- (iii) Find the velocity after 20 seconds (1)
- (iv) Find the maximum possible velocity (1)
- b) A cable car (C) is travelling at a constant distance of 100m above the ground. An observer on the ground at Point A sees the cable car on a bearing of 345° from A with an angle of elevation of 65° . After one minute the cable car has a bearing of 025° from point A and a new angle of elevation of 69° .
- Find (i) the distance the cable car has travelled in that minute. (2)
- (ii) its speed in metres per second. (1)
- c) The acceleration of a particle is given by $\ddot{x} = -4x \text{ ms}^{-2}$
- (i) Show that $x = \sin 2t$ is a solution of this differential equation (1)
- (ii) Find the times when the displacement will be zero. (1)
- (iii) Find the velocity of the particle at these times. (1)
- (iv) Hence or otherwise find the equation of its velocity in terms of its displacement. (1)

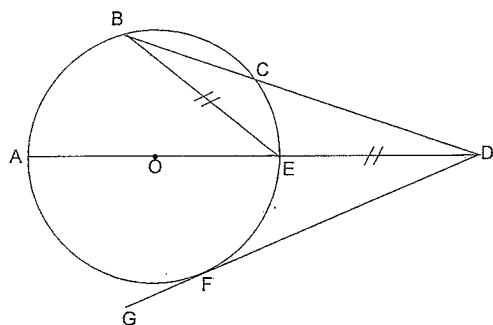
Marks

Question 4 (12 marks)

- a) Taking a first approximation of $x = 0.6$ radians, use 1 application of Newton's method to solve $\tan x = x$ correct to 2 decimal places

(2)

b)



O is the centre of a circle through points A, B, C, E and F.

Diameter AOE is produced to point D such that $BE = ED$ as shown.

- (i) Copy this diagram onto your answer page (1)
- (ii) Show that $\angle BEO = 2 \angle CDE$ (2)
- (iii) Show that $\angle BAO = 90^\circ - \angle BEO$ (2)
- (iv) DFG is a tangent to the circle touching at F. Given that $BE = 5\text{cm}$ and radius of the circle is 3.5cm , find the length of DF in exact form. (2)
- c) Solve $\frac{3}{x-4} < 5$ (3)

Marks

Question 5 (12 marks)

- a) Show that $\tan^{-1} \frac{4}{5} + \tan^{-1} \frac{5}{4} = \frac{\pi}{2}$ (2)

- b) Simplify $\frac{|x+1|}{x^2-1}$ for $x \neq \pm 1$ (2)

- c) The remainder when polynomial

$P(x) = ax^4 + bx^3 + 15x^2 + 9x + 2$ is divided by $(x-2)$ is 216 and $(x+1)$ is a factor of $P(x)$. Find "a" and "b". (3)

- d) Points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$.

- (i) Find the equation of the chord PQ. (1)

- (ii) If chord PQ subtends a right angle at the origin show that $pq = -4$. (2)

- (iii) Find the equation of the locus of the midpoint of chord PQ. (2)

Marks

Question 6 (12 marks)

- a) Use the principle of mathematical induction to prove that

$$2^n - 1 > 5n + 2 \text{ for } n > 4 \quad \text{where } n = \text{integer} \quad (3)$$

- b) (i) Find the point of intersection of the curves $y = x^2$ and $y = (x-2)^2$. (1)

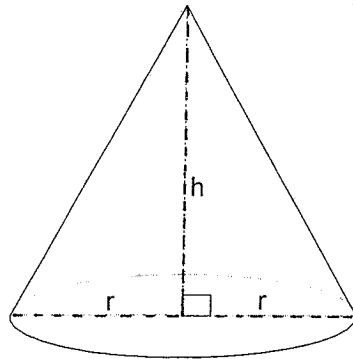
- (ii) The area enclosed by the two curves and the x axis from $x = 0$ to $x = 2$ is rotated around the y axis. Find the volume of revolution in exact form. (3)

- c) Use the substitution

$u^2 = 1 + x^3$ to evaluate correct to 3 decimal places

$$\int_0^1 \frac{3x^2 dx}{2\sqrt{1+x^3}} \quad (2)$$

- d) A funnel is dropping sand at a constant rate of 8m^3 per minute. The sand pile is in the shape of a cone such that the height (h) metres is always twice the radius (r) metres.



Find the rate at which the height is changing when the sand pile is at a height of 2m. Answer correct to 2 decimal places.

(3)

Marks

Question 7 (12 marks)

- a) (i) Find the inverse function $f^{-1}(x)$ in terms of x for $y = f(x) = x^2 - 4x$ over the restricted domain $x \geq 2$. (2)

Write domain and range of this inverse function

- (ii) Hence find the point common to both $f(x)$ and $f^{-1}(x)$ in this domain. (1)

- b) Water is flowing at a horizontal velocity of 2ms^{-1} over a 5m vertical cliff.

- (i) How far from the base of the cliff will it fall? (2)

- (ii) At what velocity will it strike the river below? (correct to 1 dec place) (1)

- (iii) At what acute angle will it make with the river below? (Use $g = 10\text{ms}^{-2}$) (to the nearest degree) (1)

- c) A particle moves in a straight line, its position x metres at time t seconds being given by $x = \sqrt{3} \sin t - \cos t$

- (i) Express $\sqrt{3} \sin t - \cos t$ in the form $A \sin(t - \alpha)$ where α is in radians and $A > 0$ (2)

- (ii) Show that the particle is undergoing Simple Harmonic Motion about $x = 0$ (1)

- (iii) Find the amplitude and period of the motion. (1)

- (iv) At what time does the particle first reach its maximum speed after time $t = 0$ secs? (1)

S.G.H.S. Trial H.S.C Maths 2006 Ext 1

Q1 a $y = 4 \cos^{-1} 3x$
 $\frac{dy}{dx} = \frac{-12}{\sqrt{1-9x^2}}$

(2)

b $x_1, y_1 = 3, 4$
 $x_2, y_2 = -5, 1$
 (externally) $k_1, k_2 = -2, 3$

$x = \frac{k_1 x_2 + k_2 x_1}{k_1 + k_2}$
 $= \frac{10 + 9}{1} = 19$

$y = \frac{k_1 y_2 + k_2 y_1}{k_1 + k_2}$
 $= \frac{-2 + 12}{1} = 10$

$\therefore (x, y) = (19, 10)$

(2)

c $\int \frac{dx}{4 + 3x^2}$

$= \int \frac{dx}{3(\frac{4}{3} + x^2)}$

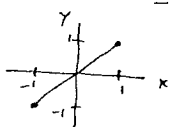
$= \frac{1}{3} \int \frac{dx}{(\frac{2}{\sqrt{3}})^2 + x^2}$

$= \frac{1}{3} \times \frac{\sqrt{3}}{2} \tan^{-1} \left(\frac{\sqrt{3}x}{2} \right) + c$

$= \frac{\sqrt{3}}{6} \tan^{-1} \left(\frac{\sqrt{3}x}{2} \right) + c$

(2)

d $y = \sin(\sin^{-1} u)$
 $\therefore y = x \quad -1 \leq x \leq 1$
 $-1 \leq y \leq 1$



(3)

e $\int 2x e^{x^2-3} dx$

$u = x^2 - 3$

$\frac{du}{dx} = 2x$

$\therefore du = 2x dx$

$\int 2x e^{x^2-3} dx$

$= \int e^u du$

$= e^u + c$

$= e^{x^2-3} + c$

(3)

Question 2

a $y = 2x \quad m_1 = 2$

$5x + 2y - 3 = 0 \quad m_2 = -\frac{5}{2}$

$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$
 $= \frac{2 + \frac{5}{2}}{1 + 2(-\frac{5}{2})}$
 $= \frac{\frac{4+5}{2}}{1-5}$
 $= \frac{\frac{9}{2}}{-4}$

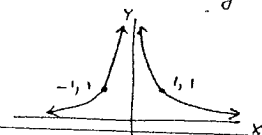
Obtuse $\theta = 131^\circ 38'$

(2)

b $y = \frac{|x|}{x^2}$

when $x < 0 \quad y = \frac{-x}{x^2} = -\frac{1}{x}$

when $x > 0 \quad y = \frac{x}{x^2} = \frac{1}{x}$



(2)

c $\int_0^{\pi/6} \cot x dx$

$= \int_0^{\pi/6} \frac{\cos x}{\sin x} dx$

$= \left[\log_e \sin x \right]_0^{\pi/6}$

$= \log_e \sin \frac{\pi}{6} - \log_e \sin 0$

$= \log_e \frac{1}{2} - \log_e 0$ (undefined number)

$= \log_e 1 - \log_e 2 - \text{undefined number}$
 $= -\log_e 2 - \text{undefined number}$

(3)

(3)

a $y = \sin^{-1} 3x$
 $\frac{dy}{dx} = \frac{3}{\sqrt{1-9x^2}}$

$x=0, \frac{dy}{dx} = 3$

$\therefore$ Gradient = $-\frac{1}{3}$

$x=0, y = \sin^{-1} 0 = 0$

$y=0 = -\frac{1}{3}(x-0)$

$\therefore 3y = -x$

$\therefore x + 3y = 0$ is normal

d $\tan 2\theta = \sqrt{3}$

$2\theta = n\pi + \tan^{-1} \sqrt{3}$

$= n\pi + \frac{\pi}{3}$

$\therefore \theta = \frac{n\pi}{2} + \frac{\pi}{6}$

where $n = \text{any integer}$

($= 0, \pm 1, \pm 2, \dots$)

(2)

Q35

$$\begin{aligned} \text{i} \quad \frac{dv}{dt} &= -k(v-C) \\ v &= C + Ae^{-kt} \\ \text{L.H.S} &= \frac{dv}{dt} = 0 - kAe^{-kt} \\ \text{R.H.S} &= -k(v-C) \\ &= -k(C + Ae^{-kt} - C) \\ &= -kAe^{-kt} \quad (1) \\ \text{L.H.S} &= \text{R.H.S} \\ \therefore v &= C + Ae^{-kt} \text{ is a soln.} \end{aligned}$$

$$\begin{aligned} \text{ii} \quad t &= 0, v = 0, C = 500 \\ \therefore 0 &= 500 + Ae^{-0} \\ \therefore A &= -500 \\ t &= 5, v = 21 \\ \therefore 21 &= 500 + 500e^{-5k} \\ 500e^{-5k} &= 479 \quad (2) \\ e^{-5k} &= \frac{479}{500} \\ k &= \log_e\left(\frac{479}{500}\right) \div (-5) \\ \therefore k &= 0.0085815 \end{aligned}$$

$$\begin{aligned} \text{iii} \quad t &= 20 \\ v &= 500 - 500e^{-20 \times 0.0085815} \\ &= 78.85 \text{ m s}^{-1} \quad (1) \end{aligned}$$

$$\begin{aligned} \text{iv} \quad v &= 500 - \frac{500}{e^{kt}} \\ \text{Let } t \rightarrow \infty, v &= 500 - 0 \\ &= 500 \text{ m s}^{-1} \\ &= \text{maximum vel} \quad (1) \end{aligned}$$

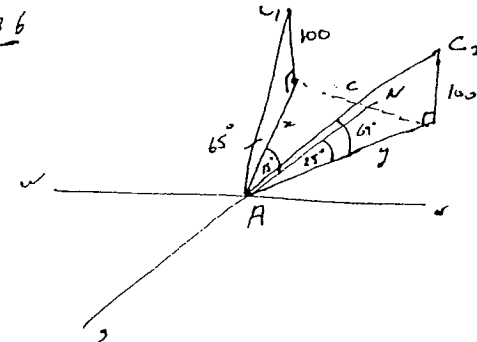
$$\begin{aligned} \text{Q36} \quad \ddot{x} &= -4x \\ \text{i} \quad x &= \sin 2t \\ \dot{x} &= 2 \cos 2t \\ \ddot{x} &= -4 \sin 2t \quad (1) \\ &= -4x \\ \therefore x &= \sin 2t \text{ is a soln.} \end{aligned}$$

$$\begin{aligned} \text{ii} \quad \text{Let } x &= 0 \\ \sin 2t &= 0 \quad (1) \\ 2t &= 0, \pi, 2\pi, 3\pi \text{ etc.} \\ \therefore t &= 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2} \text{ etc.} \end{aligned}$$

$$\begin{aligned} \text{iii} \quad \dot{x} &= 2 \cos 2t \\ t=0, \dot{x} &= 2 \text{ m s}^{-1} \\ t=\frac{\pi}{2}, \dot{x} &= -2 \text{ m s}^{-1} \\ t=\pi, \dot{x} &= 2 \text{ m s}^{-1} \quad (1) \\ t=\frac{3\pi}{2}, \dot{x} &= -2 \text{ m s}^{-1} \end{aligned}$$

$$\begin{aligned} \text{iv} \quad \ddot{x} &= -4x \\ \frac{d}{dx}\left(\frac{1}{2}v^2\right) &= -4x \\ \frac{1}{2}v^2 &= \int -4x \, dx \\ &= -\frac{4x^2}{2} + C \\ \frac{1}{2}v^2 &= -2x^2 + C \\ \text{Now } v &= 2 \text{ when } x=0 \\ \frac{1}{2} \times 4 &= 0 + C \\ C &= 2 \\ \frac{1}{2}v^2 &= -2x^2 + 2 \\ v^2 &= 4 - 4x^2 \quad (1) \\ v &= \pm \sqrt{4 - 4x^2} \\ \therefore v &= \pm 2 \sqrt{1-x^2} \text{ m s}^{-1} \end{aligned}$$

Q36



$$\begin{aligned} \text{i} \quad \tan 65^\circ &= \frac{100}{x} \\ x &= \frac{100}{\tan 65^\circ} = 46.63 \end{aligned}$$

$$\begin{aligned} \tan 69^\circ &= \frac{100}{y} \\ y &= \frac{100}{\tan 69^\circ} = 38.39 \end{aligned}$$

$$\begin{aligned} c^2 &= x^2 + y^2 - 2xy \cos 40^\circ \\ &= 46.63^2 + 38.39^2 - 2 \times 46.63 \times 38.39 \times \cos 40^\circ \\ \therefore c &= 30.1 \text{ m} = \text{Dist travelled} \quad (2) \end{aligned}$$

$$\text{ii} \quad \text{Time} = \frac{30.1}{60} = 0.5 \text{ m/s} \quad (1)$$

Q6 = Prove by M.I. that

$$2^n - 1 > 5n + 2 \text{ for } n > 4$$

Step 1 Prove true for $n = 5$

$$\text{LHS} = 2^5 - 1 = 31, 5 \times 5 + 2 = 27 = \text{R.H.S.} \therefore \text{LHS} > \text{R.H.S.}$$

$\therefore$ True for $n = 5$

Step 2 Assume true for $n = k$

$$2^k - 1 > 5k + 2 \therefore 2^k - 1 - 5k - 2 > 0 \therefore 2^k > 5k + 3$$

Step 3 Prove true for $n = k+1$

$$\text{Aim: Prove } 2^{k+1} - 1 > 5(k+1) + 2 \therefore 2^{k+1} - 1 - 5k - 5 - 2 > 0$$

$$\text{aim: Prove } 2^{k+1} - 5k - 8 > 0$$

$$\begin{aligned} 2^{k+1} - 5k - 8 &= 2 \cdot 2^k - 5k - 8 \\ &> 2(5k + 3) - 5k - 8 \quad (\text{From Step 2}) \\ &= 10k + 6 - 5k - 8 \\ &= 5k - 2 \\ &> 0 \quad \text{since } k > 4 \end{aligned}$$

$\therefore$ True for $n = k+1$

Step 4 If true for $n = k$, then true for $n = k+1$

Since true for $n = 5$, then true for $n = 6$ then $n = 7$ etc

$\therefore$ By Mathematical Induction $2^n - 1 > 5n + 2$ for all integer $n > 4$

(3)

$$\text{Q7} \quad i \quad y = x^2 \quad y = (x-2)^2$$

$$x^2 = (x-2)^2$$

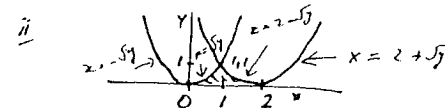
$$x^2 = x^2 - 4x + 4$$

$$4x = 4$$

$$x = 1$$

$$y = 1$$

$\therefore$ Pt of intersection = (1,1)



$$\text{Vol} = \pi \int_0^1 (4 - 4y^{\frac{1}{2}} + y) dy - \pi \int_0^1 y dy$$

$$= \pi \left[4y - \frac{4y^{\frac{3}{2}}}{\frac{3}{2}} + \frac{y^2}{2} \right]_0^1 - \pi \left[\frac{y^2}{2} \right]_0^1$$

$$= \pi \left[4 - \frac{8}{3} + \frac{1}{2} - 0 \right] - \pi \left[\frac{1}{2} - 0 \right]$$

$$= \frac{4\pi}{3} \text{ units}^2$$

(3)

$$= \int_0^1 \frac{3x^2 dx}{2\sqrt{1+x^3}}$$

Use $u^2 = 1 + x^3$

Change limits $x = 0 \quad u^2 = 1$
 $\therefore u = 1$

$x = 1, \quad u^2 = 2$
 $u = \sqrt{2}$

$$u^2 = 1 + x^3$$

$$u = (1 + x^3)^{\frac{1}{2}}$$

$$\frac{du}{dx} = \frac{1}{2} \cdot 3x^2 (1 + x^3)^{-\frac{1}{2}} = \frac{3x^2}{2\sqrt{1+x^3}}$$

$$\therefore du = \frac{3x^2 dx}{2\sqrt{1+x^3}}$$

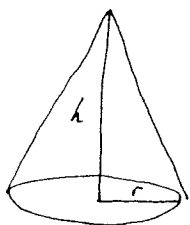
$$\therefore \int_0^1 \frac{3x^2 dx}{2\sqrt{1+x^3}} = \int_1^{\sqrt{2}} 1 du = [u]_1^{\sqrt{2}}$$

$$= \sqrt{2} - 1$$

$$= 0.414$$

(2)

86 a



$$V = \frac{1}{3} \pi r^2 h$$

$$h = 2r$$

$$\therefore r = \frac{h}{2}$$

$$r^2 = \frac{h^2}{4}$$

$$V = \frac{1}{3} \pi \frac{h^2}{4} \cdot h$$

$$= \frac{\pi}{12} h^3$$

$$\therefore V = \frac{\pi}{12} h^3$$

$$\frac{dV}{dt} = 8 \text{ m}^3/\text{min}$$

$$\frac{dV}{dh} = \frac{3\pi}{12} h^2 = \frac{\pi}{4} h^2$$

$$\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$$

$$8 = \frac{\pi}{4} h^2 \cdot \frac{dh}{dt}$$

$$\therefore \frac{dh}{dt} = \frac{32}{\pi} \cdot \frac{1}{h^2}$$

$$\text{Let } h = 2 \text{ m} \quad \therefore \frac{dh}{dt} = \frac{32}{\pi} \cdot \frac{1}{4} = \frac{8}{\pi}$$

$$= 2.55$$

$\therefore$ Rate of change of height = 2.55 m/min

47 a

$$i \quad f(x): y = x^2 - 4x \quad \text{for } x \geq 2$$

$$f^{-1}(x): x = y^2 - 4y$$

$$y^2 - 4y + 4 = x + 4$$

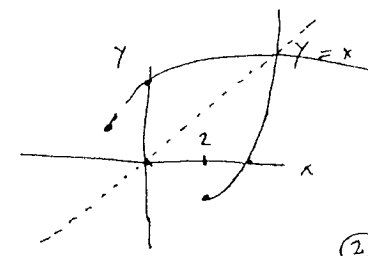
$$(y-2)^2 = x+4$$

$$y-2 = \pm \sqrt{x+4}$$

$$y = 2 \pm \sqrt{x+4}$$

$$\therefore f^{-1}(x) \text{ is } y = 2 + \sqrt{x+4}$$

$x \geq -4$ is Dom of $f^{-1}(x)$
 $y \geq 2$ is Range of $f^{-1}(x)$



ii $f(x)$ and $f^{-1}(x)$ intersect on $y = x$

$$\therefore y = x^2 - 4x = x$$

$$x^2 - 5x = 0$$

$$x(x-5) = 0$$

$$\therefore x = 5, y = 5$$

$$\therefore \text{Common Pt} = (5, 5)$$

(1)

b $\vec{x} = 2$
 $\vec{y}$

Data: $t=0, x=0, \dot{x}=2, \ddot{x}=0$

$t=0, y=5, \dot{y}=0, \ddot{y}=-g=-10$

Horizontal motion

$$\dot{x} = 2$$

$$x = 2t + c$$

$$0 = 0 + c$$

$$\therefore x = 2t$$

Vertical motion

$$\ddot{y} = -10$$

$$\dot{y} = -10t + c$$

$$0 = 0 + c$$

$$\therefore \dot{y} = -10t$$

$$y = -\frac{10t^2}{2} + c$$

$$5 = 0 + c$$

$$\therefore y = -5t^2 + 5$$

i Let $y=0$

$$0 = -5t^2 + 5$$

$$5t^2 = 5$$

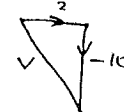
$$t^2 = 1$$

$$\therefore t = 1$$

$$x = 2 \times 1 = 2 \text{ m}$$

$\therefore$ Water falls 2m from base of cliff

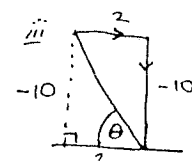
ii $\dot{x} = 2, \dot{y} = -10$



$$V^2 = 2^2 + (-10)^2$$

$$= 104$$

$$V = \sqrt{104} \approx 10.2 \text{ m/s}$$



$$\tan \theta = \frac{10}{2}$$

$$= 5$$

$$\therefore \theta = 79^\circ$$

(1)

$$Q7 \Rightarrow x = \sqrt{3} \sin t - 1 \cos t$$

$$\begin{aligned} & \perp \quad A \sin(t - \alpha) \\ & = A \sin t \cos \alpha - A \cos t \sin \alpha \\ & = (A \cos \alpha) \sin t - (A \sin \alpha) \cos t \\ & = \sqrt{3} \sin t - 1 \cos t \end{aligned}$$

$$A \cos \alpha = \sqrt{3} \quad \text{and} \quad A \sin \alpha = 1$$

$$\frac{\sin \alpha}{\cos \alpha} = \tan \alpha = \frac{1}{\sqrt{3}}$$

$$\therefore \alpha = \pi/6$$

$$A^2 \cos^2 \alpha + A^2 \sin^2 \alpha = 3 + 1 = 4$$

$$\therefore A = 2$$

$$\therefore \sqrt{3} \sin t - \cos t = 2 \sin(t - \pi/6) \quad (2)$$

ii) S.H.M. about 0 is satisfied
if $\ddot{x} = -n^2 x$ (Defn.)

$$x = 2 \sin(t - \pi/6)$$

$$\dot{x} = 2 \cos(t - \pi/6)$$

$$\ddot{x} = -2 \sin(t - \pi/6)$$

$$= -1^2 x$$

$\therefore$ S.H.M. about $x=0$ (1)

$$\text{iii) } -1 \leq \sin(t - \pi/6) \leq 1$$

$$\therefore -2 \leq 2 \sin(t - \pi/6) \leq 2$$

$$\therefore \text{Amplitude} = 2 \text{ metres}$$

$$n^2 = 1 \quad \therefore n = 1 \quad (1)$$

$$\text{Period} = \frac{2\pi}{n} = 2\pi \text{ secs.}$$

$$\text{iv) When } t=0 \quad x = 2 \sin(0 - \pi/6) \\ = -2 \times 1/2 = -1$$

Maximum speed occurs at $x=0$

$$0 = 2 \sin(t - \pi/6)$$

$$t - \pi/6 = 0 \text{ or } \pi \text{ or } 2\pi \text{ etc.}$$

$\therefore$ First reaches
max speed
when $t = \frac{\pi}{6}$ secs

(1)